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$BioH_2$ production from food waste by anaerobic membrane bioreactor

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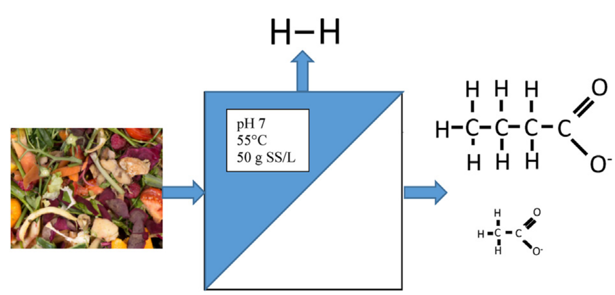
HIGHLIGHTS

- 449 mL H_2/gVS of food observed at pH 7 and at 55 °C with MBR.
- Achievement of solids free VFA with a 77% acidification ratio.
- Butyric acid pathway approved by dominancy of *Clostridium thermopalmarium* and *Clostridium butyricum*, and high B/A.
- Valorizing food waste with biorefinery approach both $bioH_2$ and VFA.

GRAPHICAL ABSTRACT



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ABSTRACT

In this study, long term continuous $bioH_2$ and volatile fatty acids (VFA) production potential in anaerobic membrane bioreactors (anMBR) from food waste by dark fermentation is investigated. Experiments of total 868 days are divided into 8 periods, each reflecting different operating conditions such as organic loading rates (OLR), hydraulic retention times (HRT) and pH regimes. Taking advantage of membrane technology, reactors are operated at solids content high as 50 g L^{-1} and reached to average 147 mL H_2/g VS added production potential. The most effective H_2 production, coupled with highest hydrolysis and acidification efficiency is achieved with pH = 7, 5 days of HRT and 18 kg COD m^{-3} d^{-1} of organic loading rate. It is also observed that in the periods which highest $bioH_2$ production was achieved, were also the periods with highest B/A ratios. Microbial community was dominated by butryic acid producers according to Denaturing Gradient Gel Electrophoresis (DGGE) and clonning analysis and it is concluded that butyric acid pathway is the dominant $bioH_2$ production mechanism in this study. With the membrane-supported completeley stirred tank reactor system, suitable conditions are not only provided for high $bioH_2$ production, but also for the production of solids free permeate with volatile fatty acids content high as 29,765 mg L^{-1}. Thus, with the innovative system proposed in this study, it is possible to establish a plant model suitable for the biorefinery concept.

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Abbreviations

anMBR	Anaerobic membrane bioreactor
B/A ratio	Butyric acid to acetic acid ratio
$bioH_2$	Biohydrogen
CSTR	Continuously stirred tank reactor
COD	Chemical oxygen demand
DGGE	Denaturating gradient gel electrophoresis
EPS	Extracellular polymeric substance
HRT	Hydraulic retention time
MBR	membrane bioreactor
MLSS	Mixed liquor suspended solids
MLVSS	Mixed liquor volatile suspended solids
OLR	Organic loading rate
ORP	Oxidation – reduction potential
PCR	Polymerase chain reaction
S_f	Net VFA production as COD equivalent
S_i	Influent soluble COD
TCD	Thermal conductivity detector
TKN	Total kjeldahl nitrogen
TS	Total solids
UF	Ultrafiltration
VFA	Volatile fatty acid
VS	Volatile solids

Introduction

Hydrogen (H_2) production is an attractive option to meet future energy demands due to its high energy intensity and low pollutant production [1]. H_2 produces only water vapor when burned and possesses the highest energy content per grams (143.35 kJ/g) [2,3] among the all fuels known to this day. Although variety of H_2 production techniques as energy source exist, using waste streams possesses a significant advantage in terms of production costs, reducing greenhouse gas emissions and waste valorization [4]. Photo-fermentation and dark fermentation are among the most widespread $bioH_2$ production methods. While photo-fermentation is a light dependent process, in dark fermentation there is no light demand, which offers a flexibility in reactor design and configuration. Besides, fermentative microorganisms in the absence of light, can produce organic acids and alchohols which can further be valorized as energy source [1,5]. Although dark fermentation appears to be advantageous, resources containing dense substrates are needed to produce significant amounts of $bioH_2$. Among high strength substrates, food waste is a very attractive solution due to its high production amount and its widespread source. This waste in turn has huge potential for valorization and energy recovery. On the other hand, inclusions of food waste have different $bioH_2$ production potentials [6] and for commercial scale applications, it is probable that composition of incoming food waste will have fluctuations. While carbohydrate – rich food waste is reported to be the most suitable for $bioH_2$ production [6] protein and fat rich wastes possess less $bioH_2$ production potential [6–8]. Furthermore, cellulosic substances, which are abundant in food preparation left overs need to be broken down into smaller units before being fermented [6].

Several reactor configurations for anaerobic $bioH_2$ production are reported in the literature [1,9–11]. While completely stirred tank reactors (CSTR)s are most frequently used [12,13], membrane bioreactor (MBR) technologies possess significant advantages comparing to other systems, such as allowing high solids content, having smaller footprint, disallowing cell washout, allowing excellent separation of solid material and metabolites, more robust control of HRT and sludge age [1,4]. Lee et al. [14], investigated the energy balance of CH_4 production by anMBR and reported positive energy yields at HRTs ranging from 5 to 50 days. Main drawback for membrane based systems, on the other hand is membrane fouling problem, which is frequently encountered. In addition to the reactor type, suitable operating conditions should also be selected for high and stable $bioH_2$ production. Anaerobic $bioH_2$ production is reported to be affected by many factors such as loading rates, HRT, pH, mixing, nutrient availability and temperature [14].

For continuous $bioH_2$ production, assessment of conditions for high production efficiency, as well as valorization of other products with biorefinery approach should be the main focus. Two stage fermentation process, in which effluent of $bioH_2$ production process is converted to CH_4 is reported to be benefical from energy recovery standpoint [6], while combining dark fermentation with photo-fermentation is also suggested [15]. On the other hand, techno-economical feasibility of co-production of $bioH_2$ and VFAs using anMBRs is clearly demonstrated by several authors [16–20]. Furthermore, for commercialization of $bioH_2$ production sytems, stability of dark fermantation should be improved [21]. Instabilities in $bioH_2$ production for continuously operated systems are frequently reported in the literature [22–25], which makes long term observation of $bioH_2$ production trend is a necessity, while taking into account the fluctuations in the feedstock characteristics. These fluctuations are known to influence $bioH_2$ production [22]. Among the studies in the literature in which food waste was used as feedstock, Chu, et al. [26], investigated two stage H_2 and CH_4 production from food waste for over 150 days. While fluctuations of feedstock characteristics were taken into account in this study, HRT and pH was kept constant at 1.3 days and 5.5 respectively. Likewise, Alpagani, et al. [27], operated $bioH_2$ CSTR for 251 days at constant HRT and pH; 5 days and 5.5 respectively, to achieve $bioH_2$ production of 104.5 mL/gVS_{added}. Other long term studies for more than 100 days for $bioH_2$ production, mostly either used other feedstock material or did not consider fluctuating inputs [28–31]. In this scope, aim of this study was to evaluate the effects of operating conditions of anMBR to $bioH_2$ production on long term, with taking fluctuations in feedstock characteristics into account, to determine optimum operating conditions for continuously operated systems, while valorizing other microbial products. For this purpose, along with continuous $bioH_2$ production, obtaining VFA rich permeate in the MBR effluent was among the objectives of this study. Experiments were divided into 8 periods of total 868 days, each reflecting different pH regimes and HRTs. Solids content were kept high as 50 g L^{-1}. As readily biodegradable food waste is produced in large amounts with high solids contents, MBR technology poses an obvious advantage for this case. Furthermore, understanding the dominant production pathway governing the bioH_2 production is of paramount importance to develop correct fermentation strategy in long term. Results are presented with DGGE analysis; microbial community and physiolocical characterization were also correlated to reveal optimum condition for obtaining both bioH_2 and VFA production. In order to increase the benefit obtained from bioH_2 production, an innovative approach with a biorefinery production model has been presented in this study.

Materials and methods

Feedstock

Food waste was collected from Istanbul Technical University's dining hall every week and stored at 4 °C prior to daily feeding through grinder. The feedstock characteristics were different regarding to menu of the day, therefore collected wastes varied in compostion. It mostly consisted of food preparation left overs such as stovers and peelings, as well as rice, grains, pasta and vegetables. After grinding food wastes by garbage disposer, they were screened through a sieve having the openings of 3 mm. All grounded food wastes passed through the sieves more than 95% of the time.

Inoculum

The bioH_2 MBR was initially loaded with a biomass characterized by Mixed Liquor Suspended Solids (MLSS) and Mixed Liquor Suspended Volatile Solids (MLVSS) concentrations of around 12.5 and 11 g L^{-1}, respectively. Inoculum for bioH_2 MBR was provided from the reactor which was previously operated to produce bioH_2 from food waste at 70 °C. Reasonable amount of bioH_2, 65–80 mL H_2/g VS_{added}, was observed at this origin of the inoculum.

Reactor setup and operation

The bioH_2 MBR was mainly composed of two complementary units, continuosly stirred tank reactor and membrane modules, sequentially (Fig. 1). Automated laboratory scale reactor with an effective total volume 11.5 L was operated at a thermophilic temperature of 55 °C 2 N HCl and 6 N NaOH were used to keep constant and desired pH. The bioH_2 MBR was operated for 868 days at which 8 periods were defined. Operating characteristics were given in Table 1. The initial aim was to keep the pH around neutral level and gradually increase the organic loading rate by keeping the substrate as received and decreasing the HRTs from 25.3 to 2.5 days. In doing so, the organic loading rate started averagely from 4.8 and increased up to around 27 g COD L−1 day−1, corresponded to period P1 to P4. Later on, low pH (5) and variable pH (5–8) operations were carried out for the period of P7 and P8, respectively.

The flows of the reactor units were controlled by the rate controlled peristaltic pump that controls the wastewater according to the desired set value by level switches. Oxidation and reduction potentials were also measured to control the anaerobic environment, if there was any leakage in the reactor units.

MBR system was composed of submerged UF membrane module with a 2 mm fiber outer diameter, 49 cm fiber length and 0.05 μm pore size. Tap water flow was 180 L m^{-2} h^{-1} (−700 mbar, 25 °C). Nitrogen gas sparging, backwashing with permeate, and chemical wash were the techniques used for Membrane cleaning in this study. On-site nitrogen gas generator was used for the gas sparging period. Nitrogen gas was fed every hour by sparging 1 min with a flow rate around 5.8 L min^{-1}. The permeate collected from the submerged membrane transferred to the discharge tank. During this transfer, the pressure on the suction line was also measured and recorded. In between the days 159 and 202, the reactors were run but analytical parameters were not measured due to sampling problems. The gas amount and composition were also not measured due to flaw in gas collection system at the time intervals of 181–226, 473–560, and 607–680. Therefore, the missing data were not shown in related graphics.

COD solubilization and acidification ratios

COD (chemical oxygen demand) Solubilization ratio indicates the stability and efficiency of VFA production. It was calculated by the following equation:

COD solubilization (%) = [(Effluent soluble COD − Influent soluble COD) / Influent COD] x 100 [1]

Acidification ratio was the critical parameter to figure out the fermentation efficiency. It was calculated by the following equation:

$$\text{Acidification ratio} = (S_f / S_i) \times 100 \quad [2]$$

S_i denotes influent soluble COD concentration; S_f denotes net VFA production as COD equivalent.

Analytical methods

Suspended Solids (SS), Volatile Solids (VSS), MLSS, MLVSS, alkalinity, COD, TKN, ammonia, were analyzed according to Standard Methods [16]. The pH measurement was carried out with a pH-meter (Thermo Orion 720 A, Mansfield, TX, USA). Conductivity was determined by using potable conductivity meter (Hach Sension 5, Loveland, CO, USA). Samples, which were taken for soluble COD and VFA were centrifuged at 9000 rpm for 15 min and the resulting supernatant was filtrated through a Millipore PVDF filter (0.45 mm) for COD and Millipore PVDF filter (0.22 mm) for VFA. COD samples were preserved with H_2SO_4, VFA samples with 10 M H_3PO_4.

The VFA levels were determined by a gas chromatograph (Shimadzu GC-2010, Kyoto, Japan) equipped with a flame-ionisation detector and a 30 m × 0.25 mm TRB-FFAP capillary column (film thickness = 0.25 μm). The temperature of the injection port and detector were 250 °C. The oven temperature reached 60 °C in first 1 min and then 60 °C–230 °C (5 °C/min) and fixed at 230 °C in 1 min. Nitrogen was the carrier gas at 30 mL/min. In addition, H_2 gas was used at 40 mL/min flow rate and air flow was used at 400 mL/min. The sample (1.0 mL) was transferred into a gas chromatography vial to which 0.2 mL of 10% phosphoric acid was added. H_2, CO_2, CH_4

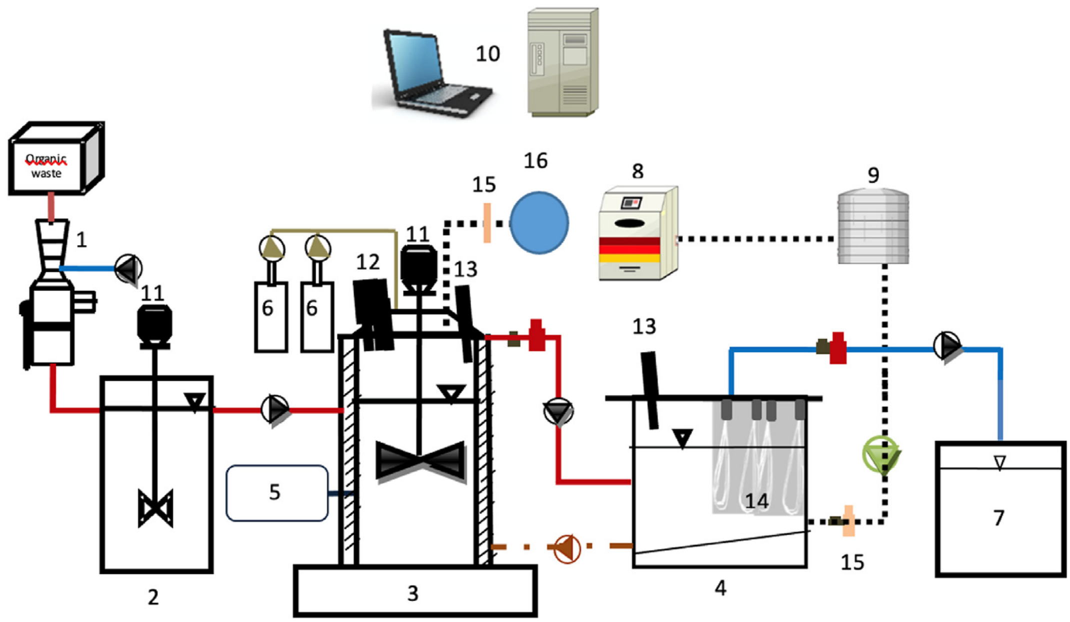

Fig. 1 – Experimental set up of lab-scale bioH_2 AnMBR. 1-Garbage disposal unit; 2-Feed tank; 3-BioH_2 reactor; 4-Membrane module; 5-Hot water circulation system; 6-Acid, base dosage system; 7-Permeate tank; 8-Nitrogen gas generator; 9-Gas chamber; 10-PLC; 11-Top entry mixer; 12-Temperature, pH and ORP sensors, 13-Level sensors; 14-Submerged PVDF membranes, 15-Gas Flowmeter, 16, Gas storage bag.

contents of the biogas were measured with a gas chromatograph (Perichrom PR2100, France) equipped with a thermal conductivity detector (TCD) and helium and nitrogen served as the carrier gases. Atomic absorption spectroscopy method (Analytikjena, Germany) was used to determine metal contents of the reactors. Some anions (chloride, bromide, fluoride, sulphate, nitrate, nitrite, and phosphate) were determined by using an ion chromatograph (Dionex ICS-3000).

Molecular methods and DGGE analysis

DNA extraction from biomass samples, taken at a different time point of operation, were accomplished by using ZR Soil Micobe DNA Kit (ZymoResearch, Irvine, CA, USA) after mechanical disruption by bead beating. After verifying the concentration and purity of DNAs, DNA samples were stored at −20 °C until the DGGE-PCR experiments. PCR was performed using forward primer U968-GC with 40-base GC-clamp (5′-CGC CCG GGG CGC GCC CCG GGC GGG GCG GGG GCA CGG GGG GAA CGC GAA GAA CCT TAC-3′) and reverse primer L1401 (5′-GCG TGT GTA CAA GAC CC-3′) to amplify V6–V8 variable region of 16S rDNA. PCR ingredient was consisted of dNTPs (0.2 mM each), primer pair (10 pmol of each), 1.25 unit of DNA Taq polymerase, and 1X PCR buffer including 20 mM $MgCl_2$. PCR conditions were as follow; the initial denaturation at 94 °C for 5 min, 35 cycles at 94 °C for 30 s for denaturation, 60 °C for 1 min for annealing and 72 °C for 1.5 min for extension. A final extension phase was performed for 5 min at 72 °C. DGGE was performed by using Dcode System (BioRad, Hercules, CA, USA). The GC clamp PCR-products were separated in 8% (w/v) polyacrylamide gels containing a linear 35–58% denaturant gradient (80% denaturant corresponds to 7 M urea and 40% deionized formamide). Electrophoresis was performed at 85 V and 60 °C for 16 h in 1 X TAE (Tris-acetate-EDTA). Gels were stained using silver nitrate and scanned at GS-800TM Calibrated Densitometer (Bio-Rad, Hercules, CA, USA). DGGE images were normalized and dendrogram analysis were performed by using GelCompar II Software version 6.5 (Applied Maths, Kortrijk, Belgium). Cluster analysis of bacterial community were performed using an unweighted pair

Table 1 – Operating characteristics of bioH_2 MBR.

Period	Duration of period [days]	pH	HRT [days]	Organic Loading Rate	
				[g COD L^{-1} day^{-1}]	[g VS L^{-1} day^{-1}]
P1	268	7	1.2–25.3	4.58 ± 5.63	2,33 ± 3,25
P2	76	7	10	8.80 ± 1.07	5,64 ± 2,38
P3	266	7	5	18.61 ± 4.36	8.07 ± 3.47
P4	97	7	2.5	26.68 ± 3.39	18.77 ± 8.41
P5	63	8	5	12.72 ± 0.99	5.23 ± 1.93
P6	42	6	5	12.47 ± 0.38	5.10 ± 0.93
P7	28	5.5	5	11.96 ± 0.89	5.62 ± 0.77
P8	28	5–8	5	10.88 ± 0.48	5.86 ± 1.52

group method with arithmetic mean (UPGMA). Samples showing different DGGE profile were selected for cloning and sequencing. Amplified 16S rRNA regions with forward primer BAC-968F (5′-AAC GCG AAG AAC CTT AC-3′) and reverse primers BAC-1401R (5′-GCG TGT GTA CAA GAC CC-3′), BAC-1492R (5′-CGG CTA CCT TGT TAC GAC-3′) were purified by High Pure PCR Product Purification Kit (Roche, Mannheim, Germany). Purified amplicons were cloned into pJET1.2 (blunt-end cloning vector) by CloneJET™ PCR Cloning Kit (Thermo Scientific) and the ligation products were transformed to *E. coli* Top 10 (Invitrogen by Life Technologies) and JM107 (MBI Fermentas) cells via TransformAid Bacterial Transformation Kit (Thermo Scientific). Colony PCR was performed to confirm the presence of insert and 48 clones with insert were screened by DGGE to determine samples to be sequenced. Sequence similarity was determined by using Basic Local Alignment Search Tool program (BLAST, https://blast.ncbi.nlm.nih.gov/Blast.cgi). Phylogenetic tree was constructed using the neighbor-joining method by MEGA Version 10.1.8.

Results and discussion

Operation of BioH$_2$ AnMBR

There is no consensus on the literature for optimum loading rate for bioH$_2$ production; some authors reported that high loading rates may decrease bioH$_2$ production [32], while some others reported the opposite [33–35], supporting the opinion that optimal conditions for continuous bioH$_2$ production is dependent on many other factors along with loading rates, such as fermentation products, feedstock characterization, pH, HRT, etc.

pH is among the factors that affect bioH$_2$ production since it directly influences the activity of hydrogenase enzyme. According to the general opinion, high pH values generally cause low bioH$_2$ production. For instance, Shin, et al. [36], reported decrease in the H$_2$ content of biogas when pH was increased to 6.5 from 4.5, while lag phase was longer for pH = 4.5. Liu et al. [37], found optimum pH conditions could be maintained by a controlled decrease from 6.6 to 4. Van Ginkel et al. [38], studied effect of varying pH values from 4.5 to 7.5 on bioH$_2$ production and reported pH value of 5.5 to be optimum. Contrary to this; Liu et al. [39], recorded higher bioH$_2$ productivity with increasing pH and found pH = 7 to be optimum. They also hypothesized that minimum pH value might be 4.7 for thermophilic bioH$_2$ production. Furthermore, biodegradation of lignocellulosic biomass, which is abundant in the feedstock, is found to be maximum near neutral pH [40], which is also confirmed by findings of Magnusson et al. [41], who observed bioH$_2$ production from $\alpha -$ cellulose at pH = 7. In addition, having VFAs in ionic form at high pH conditions will reduce the inhibition effect, so it will be possible to obtain high products. In the light of these data, strict pH control was applied in first 4 periods of this study and pH was kept neutral at 7. For last 4 periods, pH was let free and varied between 5.5 and 8.

Taking advantage of membrane technology, reactors were operated at solids content high as 50 g L^{-1}; average influent VS concentration being 40.2 g L^{-1} and TS concentration 44.9 g L^{-1} (Table 2). As mentioned above, operating with high solids

Table 2 – Compositional analysis of bioH$_2$ MBR at 8 operating periods.

Period	Duration [day]	Flux [L m^{-2} h^{-1}]	TS [g L^{-1}]	VS [g L^{-1}]	TS/VS [%]	totCOD Inf. [mg L^{-1}]	solCOD inf. [mg L^{-1}]	totCOD Eff. [mg L^{-1}]	COD Rem. [%]	TKN inf. [mg L^{-1}]	TKN eff. [mg L^{-1}]	Ammonia Inf. [mg L^{-1}]	Ammonia Eff. [mg L^{-1}]	Ammonia/ TKN Inf. [%]	Ammonia/ TKN Eff. [%]	Conductivity., Inf. [mS cm^{-1}]	Conductivity., Eff. [mS cm^{-1}]
P1	268	2.45 ± 4.6	65.44	59.53	91	90,549	32,350	32,516	64	877	410	133 ± 171	89 ± 87	15	22	–	–
P2	76	0.82	60.54	57.85	96	88,076	42,723	46,210	48	766	415	68 ± 21	104 ± 41	9	25	–	–
P3	266	1.60 ± 0.29	46.56	40.32	87	87,702	36,384	48,107	45	1393	734	89 ± 48	269 ± 146	6	37	15	24
P4	97	1.55	65.31	58.49	90	90,509	41,483	48,767	45	2114	894	94 ± 24	304 ± 111	4	34	13	21
P5	63	0.77	28.99	26.84	93	86,115	43,860	47,895	44	981	1120	115 ± 74	633 ± 225	12	57	7	18
P6	42	0.77	30.00	25.46	85	83,674	43,194	43,952	46	809	496	122 ± 133	204 ± 128	15	41	10	17
P7	28	0.77	23.67	16.89	71	82,012	38,296	41,498	49	615	301	59 ± 15	82 ± 48	10	27	11	19
P8	28	0.77	39.33	36.33	92	74,337	33,003	38,250	49	524	320	42 ± 5	53 ± 12	8	17	11	19

NM. Not measured.

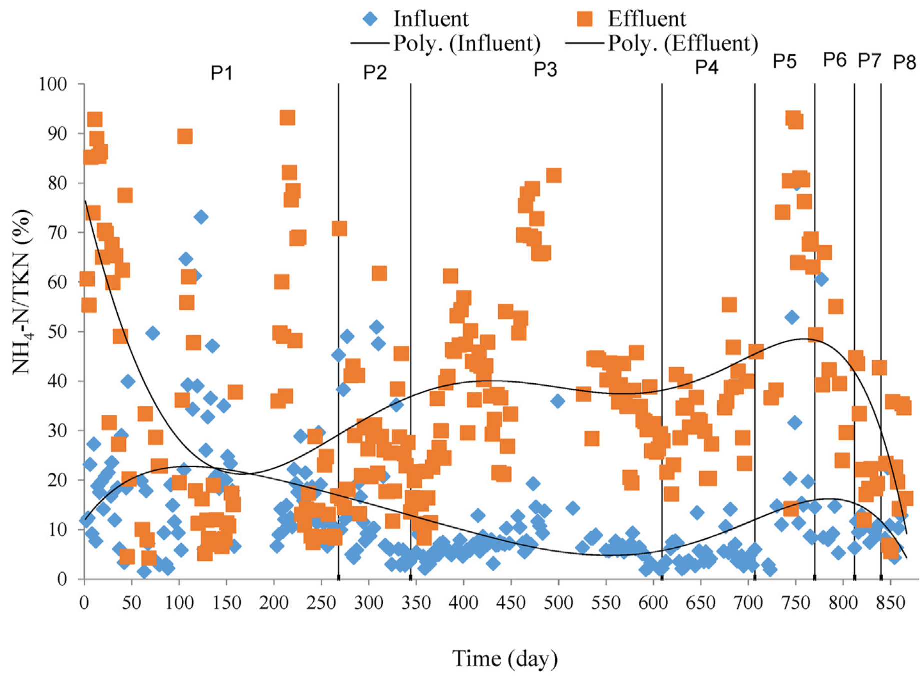

Fig. 2 – NH_4–N/TKN Ratio during each study period.

content is one of the major advantages of MBR technology. In general anMBRs are operated at solids concentrations high as 50 g L^{-1} [42]. Taking advantage of this, it has been aimed to keep solids content at 5% throughout the study. Fluctuations at solids content are due to characteristics of feed waste fed to the reactor. In the period when HRT was 2.5 days, solids concentration inside the reactor has increased to 126.33 g L^{-1} due to low HRT. In order to overcome this solids accumulation issue, decreasing 30 days sludge age could be a solution.

HRT directly influences bioH_2 and VFA production in continuous systems. Besides, it also has an effect on product composition of hydrolysis [43]. However, like in the case of loading rates, there is no consensus in literature about optimal HRT. HRTs reported in the literature for continuous bioH_2 production can be in the magnitude of hours to days; while researchers working with a well defined synthetic medium or readily biodegradable carbon source, mostly preferred HRTs in magnitude of hours HRTs in the magnitude of days are frequent for studies conducted with raw food waste [23–25,27–30,44–50]. Furthermore, Han and Shin [51] reported that acidification of organic wastes require an HRT more than 3 days. In this scope, HRTs in this study varied between 1.2 and 10 days during the 8 experimental periods.

Membrane was backwashed with permeate in first 270 days of the study; however this was not effective in the ultrafiltration membrane with low flux values due to high operating solids content. After 270th day, membrane was purged with nitrogen gas through membrane surface. Flux

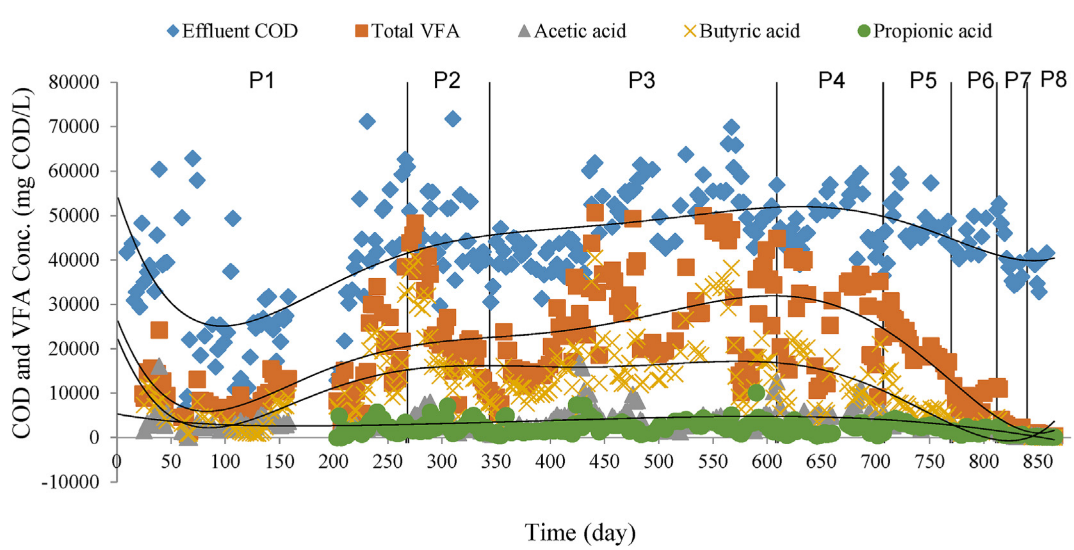

Fig. 3 – Concentration of VFAs in the effluent stream.

values varied between 0.77 and 1.60 except fort he first period where HRT varied. A 2–80 L m^{-2} h^{-1} flux values are reported in the literature for anMBRs [52]. Lee et al. [14], reported flux values from 0.6 to 1 L m^{-2} h^{-1} for an anMBR with 8.6% solids content and high loading rates from 70 to 125 kgCOD m^{-3} d^{-1}. Decreased flux values result from high solids content, besides it has been reported that thermophilic conditions boost extracellular polymeric substances (EPS) production which is among the causes of membrane fouling [52,53]. When HRT was stabilized at 5 days, a corresponding stabilization at the flux values were also observed.

As bioH_2 production was aimed at this study, methanogenic environment was prevented. Since the highest negative ORP value is observed when carbon is converted to its most reduced form, CH_4, ORP values were kept at a certain level. Very negative ORP values were observed in last 3 periods of the study (data not shown), which is in accordance with lowest bioH_2 production in these periods. However, poor bioH_2 production in these periods can be attributed to poor hydrolysis and acidification efficiency. ORP values became more negative with increased HRTs, so it should continuously be monitored.

Ammonia/TKN ratio was also monitored during the study as it is an indication of efficiency of hydrolysis (Fig. 2). Effluent ammonia/TKN ratios was higher than those of influents as the result of degradation of organic nitrogen. As free form of NH_3 is the main inhibitor of anaerobic digestion process [32,33], hydrolysis of TKN should be continuously monitored. Postor – Poquet et al. [54], investigated the effect of high solids content on free ammonia formation and observed accumulation of both NH_3 and VFAs, as solids content increased and pointed out the risk of free ammonia inhibition for systems operating with high solids content. Influent ammonia concentrations in this study ranged between 42 and 133 mg L^{-1} which is efficient in the anaerobic treatment processes [55] while operation at neutral conditions decreased risk of ammonia inhibition.

Conductivity can be defined as electricity conduction capacity of ions dissolved in the water. As a result of hydrolysis of particulate matter, ions are converted into free form in water. Thus, conductivity values of permeate were higher than those of feed in all periods. Conductivity values of permeate are observed to be high as 24 mS cm^{-1}. It can be said that permeate has characteristics of brackish water. Highest conductivity was observed 3rd period, the period in which the most efficient hydrolysis and highest bioH_2 production was observed. Furthermore it can be stated that these conductivity values didn't have negative effect on biodegradation of organic matter.

Low molecular weight ions such as Cl^-, SO_4^{2-}, K^+, Mg^{2+} and Ca^{2+} are expected to pass through UF membrane. Concentration of Cl^- has increased in first five periods, in accordance with hydrolysis of organic matter. SO_4^{2-} concentrations were generally low while removal efficiencies varied between 25% and 48% for the periods 1, 5 and 6. For the 8th period, SO_4^{2-} concentration did not change considerably. SO_4^{2+} concentration is expected to increase with hydrolysis of particulate matter and decrease due to activity of SO_4^{2+} reducers. Since these activities happen simultaneously in the same reactor, it is difficult to assess which reaction is dominant in which operating period. However, in the 3rd period with the most

Table 3 – Compositional analysis of bioH_2 MBR at 8 operating periods.

Period	Acetate[a] Eff. [mg L^{-1}]	Aceate[a] Eff./totCOD Eff. [%]	Propionate[a], Eff. [mg L^{-1}]	Propionate[a] Eff./totCOD Eff. [%]	Butyrate[a], Eff. [mg L^{-1}]	Butyrate[a] Eff./totCOD Eff. [%]	totVFA[a] [mg L^{-1}]	totVFA[a] Eff./totCOD Eff. [%]	Acidif. Ratio [%]	Solubil. Ratio [%]	B/A ratio (g/g)	Sulphate, Inf. [mg L^{-1}]	Sulphate, Eff. [mg L^{-1}]
P1	3131	10	2111	6	7668	24	12,161	37	39	−0.3	2.9	68	37
P2	3330	7	2741	6	19,066	41	25,437	55	56	3	6.1	83	96
P3	4174	9	2506	5	17,932	37	29,765	62	77	13	5.3	50	198
P4	5432	11	1831	4	11,722	24	27,177	56	64	8	2.7	40	57
P5	3337	7	2740	6	7733	16	21,523	45	43	4	2.5	60	43
P6	1821	4	1417	3	2170	5	7497	17	14	−3	1.3	104	60
P7	797	2	936	2	1446	3	4126	10	8	5	1.9	47	52
P8	358	1	182	0	49	0,1	661	2	1	−6	0.1	43	44

[a] Expressed as COD equivalence. NS. No solubilization. NM:Not measured.

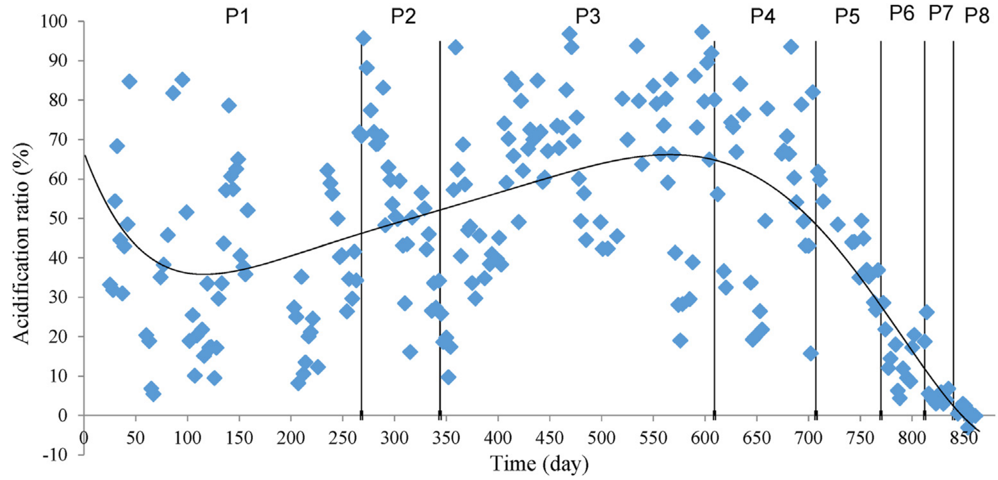

Fig. 4 – Acidification ratios for each study period.

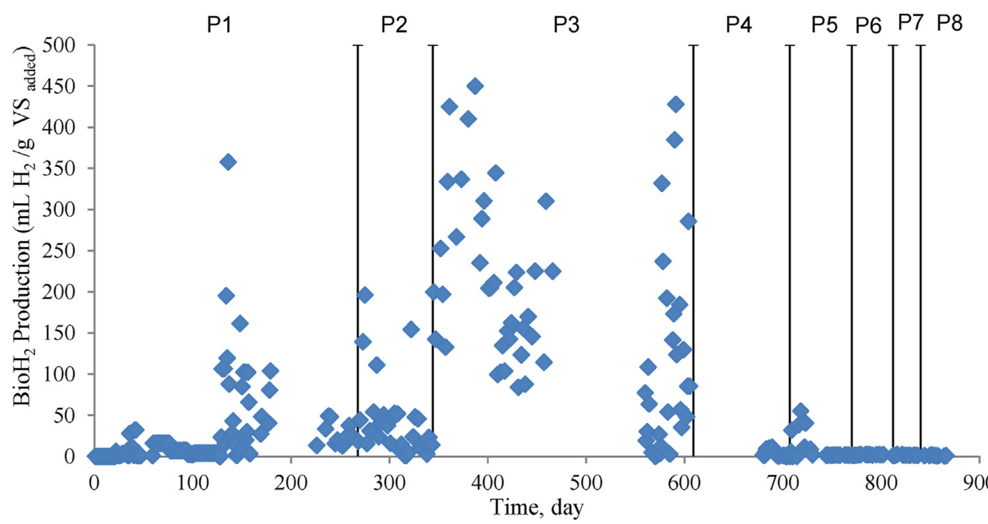

Fig. 5 – $BioH_2$ production in each study period.

efficient acidification and bioH_2 production, SO_4^{+2} production increased considerably, which also indicates efficient hydrolysis.

Acidification efficiency

VFA production in the system indicates efficiency of fermentation reactions, as well as bioH_2 production. As result of fermantation, organic matter is converted to VFAs and bioH_2. As VFAs are commercially valuable products, simultaneous production of these compounds along with bioH_2 should be of interest for a more profitable full scale anaerobic facility. While CH_4 production is the main goal for most of the anaerobic processes research [56] shifting the focus towards VFA production offers number of advantages including bioreactor stability, decreased operational costs, as well as higher commercial profits [57,58].

For the 8 periods of operation, concentrations of VFAs in the effluent stream are given in Fig. 3 and Table 3. In most of the study, it was observed that VFA constitutes a significant part of the output COD. In the 3rd period, when bioH_2 production was produced the most, 62% of the effluent total COD was found to be VFA. This value showed the hydrolysis efficiency and also how high the market value of bioH_2 effluent was.

Since acidic conditions were observed in periods 6, 7 and 8, VFA production in these periods is considerably low, indicating decrease of VFA production in acidic environment. In the period 5, reactor was operated at basic environment (pH = 8). In this period 28% decrease in VFA production was

Table 4 – Average $BioH_2$ production for each study period.

Period	H_2 production, mL H_2/g VS added
P1	19.99
P2	41.87
P3	146.87
P4	3.78
P5	9.91
P6	1.86
P7	1.49
P8	0.89

Table 5 – Affiliation of the 16S rRNA V6–V8 region sequences obtained from major DGGE bands. Closest relatives were determined by using BLAST network.

Closest genera	Band no	Similarity	NCBI accession no
Acetobacter sp.	63.4	100	EU074058.1
Acetobacter peroxydans	63.3	99	AJ419836.1
Clostridium butyricum	275.1, 275.2	100	KC195777.1
Clostridium thermopalmarium	63.2, 275.3	99	NR_026112.1
Delftia sp.	4.5	99	KF059265.1
Dysgonomonas capnocytophagoides	151.1	99	AB548674.1
Leuconostoc sp. SAP81.2	4.3	95	JX067698.1
Tepidiphilus succinatimandens	4.2, 4.4, 4.6, 4.7, 4.8, 4.9, 63.1	99	NR_025725.1
Thermoanaerobacterium thermostercus	63.6	99	FM999998.1
Uncultured bacterium clone D5	4.1	99	AM500717.1
Uncultured *Bacteroides* sp. Clone MFC-B162-G01	63.5	99	FJ393126.1

observed comparing to period 3 with same HRT and pH = 7. In the period 2, with pH = 7 but higher HRT (10 days) 15% decrease for VFA production was observed comparing to period 3.

Wijekoon et al. [53], reported a positive correlation with VFA generation and loading rates. Similar pattern is observed in the present study; periods 3 and 4, being the periods with two highest loading rates, lead to highest VFA generation. For periods 1 and 2, with neutral pH conditions like periods 3 and 4, acidification ratio decreased with decreasing the loading rate, while very low acidification in last three periods can be interpreted as a result of acidic environment, as mentioned above. Acidification ratios for each period is given in Fig. 4.

Acidification ratios have been observed to be consistent with VFA production. Highest acidification ratio was observed in period 3, with 77%, which is followed by period 4 (64%) and period 2 (56%). The highest acidification efficiency was observed at period 3, which is also the period with highest bioH_2 production. This was expected as efficient bioH_2 production is directly related to acidification efficiency.

It has been reported that, among the VFAs, acetic acid has the highest market size, followed by propionic and butyric acid, while butyric acid has the highest market value [59]. Except for the last period, butyric acid is by far the most produced VFA component throughout this study.

BioH_2 production

During the study many ups and downs in the bioH_2 production was observed, as can be seen in Fig. 5. The highest and widest range of production in period 3 was between 3 and 410 mL H_2/g VS_{added}, which shows high instability. Instabilities in the bioH_2 production have been commonly reported as a typical issue in long term operations and their reasons have been discussed in the literature [21,62–65]. In the work of Lima et al. [63], which was similar to our case as no CH_4 and H_2S was observed in the gas phase; this phenomena was linked to consumption of H_2 and CO_2 by methanogenic or sulfidogenic organisms. Hawkes et al. [66], hypothesized that instabilities are caused from homoacetogenic organisms which use acetyl – CoA pathway, where CO_2 and H_2 are converted into acetic acid. Homoacetogenic bacteria belong to *Acetobacterium, Butyribacterium, Clostridium* are also observed in our microbial community (Table 5). Instable bioH_2 production can also possibly be related to variations in the daily feed characteristics. However as many reactions happen simultaneously in the system, the understanding of the reasons behind the bioH_2 production instability require further research.

Compounds known as intermediates such as acetate, propioante, butyrate, ethanol are among significant indicators

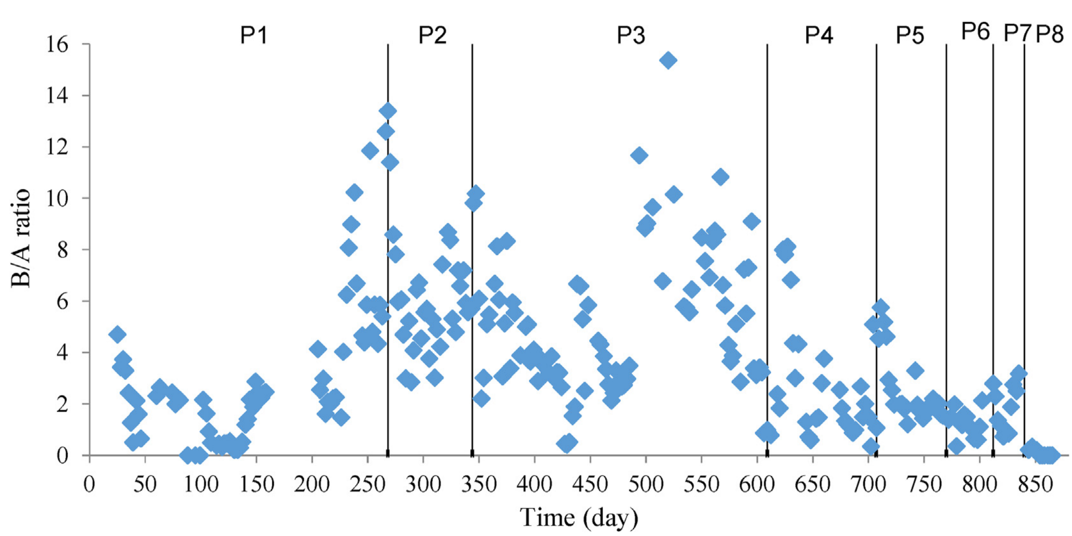

Fig. 6 – B/A ratio for each study period.

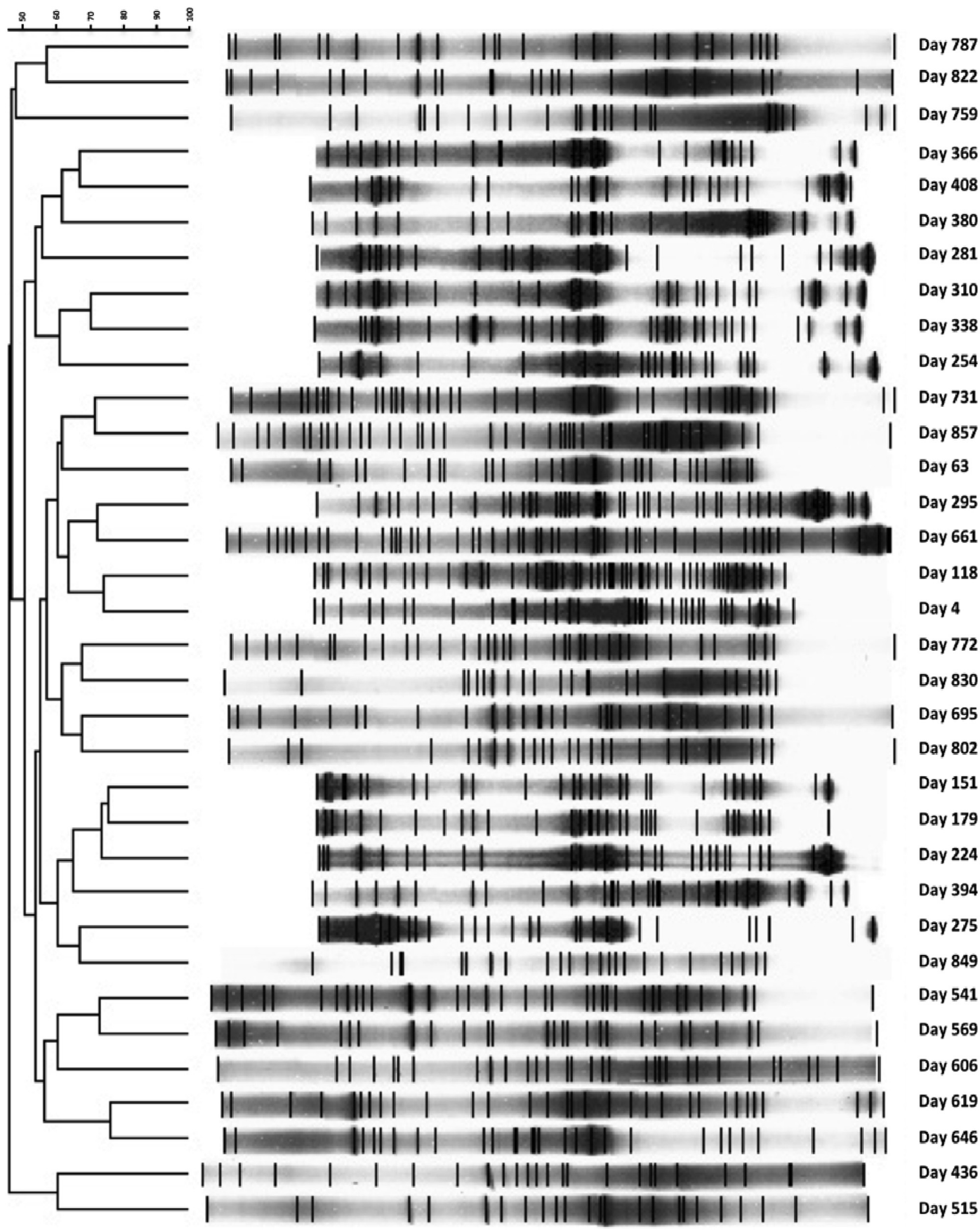

Fig. 7 – DGGE Profile and dendogram obtained by comparing 16S rRNA V6 – V8 variable region of the bacterial community in the biomass samples taken at different time points (between 4th and 857th days) Dendrogram analysis were performed using unweighted pair group method with arithmetic mean (UPGMA) and the Dice coefficient via GelCompar II Software.

of efficiency and continuity of bioH_2 production. It has been reported that acetate and bioH_2 production was higher near pH 5 [67]. Between pH range of 4–4.5 butyrate production is reported along with the bioH_2 [67]. On the other hand, theoretically most stable and efficient bioH_2 production intermediate is known to be acetate between pH values 4.5–5 [68]. Reungsang et al. [48], investigated bioH_2 production from food waste in a CSTR and they observed increase in both hydrogen yield and hydrogen production rate when HRT was decreased from 3.5 days to 2.5 days while pH was kept at 5. In our case while 5 days of HRT was found to be optimum, when HRT was decreased to 2.5 days (period 4), significantly lower bioH_2 production was observed. This pattern may be due to high ligno-cellulosic content of our feedstock which requires more time for hydrolysis, since it included considerable amount of food preparation left overs. On the other hand, B/A ratio was also considerably lower in period 4, comparing to period 3 and low bioH_2 production in this period may be attributed production of acetate from H_2 by acetogens [64] Considering acidification efficiency period 4 was high as 64%, the latter reason seems more reasonable. It is possible that contrary to low pH conditions, high amount of bioH_2 production can be achieved in neutral pH (7–7.5) [39,41]. Similarly, in our study, the potential of bioH_2 production at pH = 7 was higher than lower pHs. The highest bioH_2 production was observed in 3rd period with an average 147 mL H_2/g VS added (Table 4). Highest SO_4^{+2} production in the same period is an indicator of hydrolysis efficiency, as well as highest acidification was also observed in this period, which indicates the operating conditions for 3rd period were favorable for continuous bioH_2 production from food waste; with neutral pH and 5 days of HRT. In the 2nd period, second highest bioH_2 production was observed with 42 mL H_2/g VS added which is the also period with the highest B/A ratio. In this period HRT was increased to 10 days and acidification ratio decreased by 15% comparing to 3rd period.

Butyric acid/acetic acid ratio is an indicator of H_2 production in anaerobic systems. Since butyrate captures H_2 two times more than acetic acid [6], the effluent B/A ratio is expected to be low as possible for effective bioH_2 production. Contrary to this, highest bioH_2 production in this study is observed in period 3 where B/A ratio was also the highest as shown in Fig. 6. Reason of high butyric acid concentration in period 3 may be the consumption of acetic acid due to conversion of acetic acid to butryic acid, since they have same production pathways. However, considering effective bioH_2 production parallel with high B/A ratio, it may be more likely that acetic acid is consumed by other bacteria. Indeed, according to DGGE analysis, it is seen that microbial community was dominated by two butyric acid producing bacteria; *Clostridium thermopalmarium* (bands 63.2 and 275.3) and *Clostridium butyricum* (bands 275.1 and 275.2) (Fig. 8).

Arooj et al. [31], observed highest bioH_2 production yield at highest B/A ratio of 4.3 and suggested that along with B/A ratio, butyrate producing bacteria also governs the bioH_2 yield. Kim et al. [70], reported that B/A ratio of 4 was favorable for bioH_2 production. Furthermore, Lee, et al. [14], reported that, the optimum B/A ratio for bioH_2 production is dependent on many factors such as, substrate, culture conditions and temperature during the operation. Trad et al. [4], also reported that butyric acid pathway was the main mechanism for bioH_2 production in his study.

It is also reported that H_2 may be consumed during the production of propionate [71,72], however, this is not the case in our tests since in the period which highest bioH_2 production was observed, propionate was high as 2506 mg COD L^{-1}.

While the organic loading rate was highest in this period, this parameter is not strictly correlated with bioH_2 production rate. For example Lee et al. [14], observed 111.1 mL H_2 g^{-1} VS added at loading rate at 125.4 kg COD m^{-3} d^{-1}, which is much higher than the range of loading rates investigated in this study. Considering the B/A ratio was lower in the 4th period and butyric acid pathway being the main bioH_2 production mechanism, which is also supported by DGGE analysis, this may explain the low bioH_2 production in this period. Accordingly, B/A ratios for following periods were also lower, coupled with lower bioH_2 production. For the periods from 5 to 8, HRT

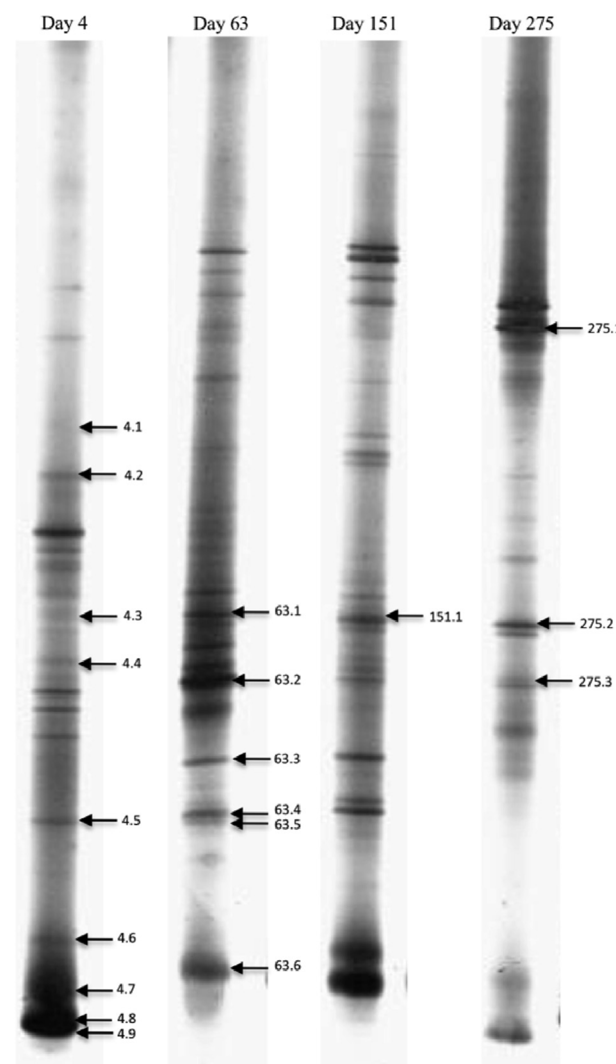

Fig. 8 – DGGE profiles of 16S rRNA V6 – V8 variable region of the bacterial community in the biomass samples taken at different time points (Day 4, 63, 151 and 275). The bands shown by arrows were sequenced after cloning of the samples with different DGGE profile.

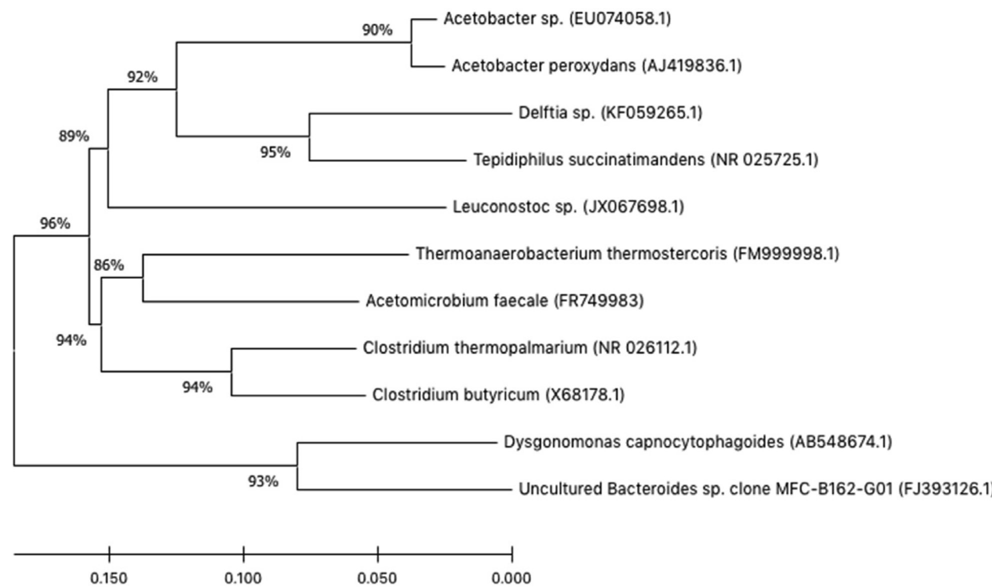

Fig. 9 – The phylogenetic tree shows the relationship between identified species in the biomass samples taken different time points. The tree was constructed from 16S rRNA V6–V8 variable region sequences of species using the neighbor-joining method by MEGA Version 10.1.8.

was kept constant, however no pH control was applied. Acidification rate was very low in these periods which is the cause of very low bioH_2 production. Results from these 4 periods supports the idea that acidic conditions are not favorable for efficient acidification.

Results of the study suggest that the main governing parameter for bioH_2 production is B/A ratio, and main bioH_2 production pathway is found to be through butyric acid (Eq. (3)), which is approved by microbial community analysis.

$$C_6H_{12}O_6 + 2H_2O => 2H_2 + CH_3CH_2CH_2COOH + 2CO_2 \quad [3]$$

Microbial shift

The microbial community of biomass samples taken at different time points were analyzed by PCR-DGGE of 16S rDNA V6–V8 variable region. The similarity of DGGE (Fig. 7) profiles was over 50% and bacterial diversity of different time points showed no significant variation.

The major bands of samples taken at different time points were identified by cloning and sequencing (Fig. 8). The sequences of major bands were aligned to closest relatives in GenBank database by BLAST network. The sequence affiliation and similarity rates are given in Table 5. By comparing DGGE profiles, it was observed that the bacterial diversity did not significantly change at different time points. *Thermoanaerobacterium thermostercus*, *Clostridium thermopalmarium*, and *Clostridium butyricum* are found to be as dominant species regarding their prominent and widespread band profiles in the DGGE pattern. Additionally, the species such as *Acetomicrobium faecale*, *Tepidiphilus succinatimandens*, *Leuconostoc* sp., *Delftia* sp., *Acetobacter* sp., *Thermoanaerobacterium thermostercus* and *Dysgonomonas capnocytophagoides* were also identified by cloning of samples with different DGGE profiles. Phylogenetic tree was constructed using the neighbor-joining method by MEGA Version 10.1.8 (Fig. 9).

Conclusions

In this study long term operating conditions are investigated for co – production of bioH_2 and VFAs from food waste in an anMBR. Experimental duration is divided in 8 period and optimum condition is found to be 5 days of HRT and pH = 7 at period 3. In this period bioH_2 production of 147 mL H_2/gVS_{added} is coupled with total VFA concentration in the effluent with 29,765 mgCOD L^{-1}. Most produced VFA was butyric acid with a concentration of 17,932 mgCOD L^{-1}. Butyric acid is the VFA with the highest market value and it is notable that its production in this amount is coupled with highest bioH_2 production.

Results of the study suggest that the main governing parameter for bioH_2 production is B/A ratio, which is also found to be correlated to microbial community. Two dominant species of microbial community, *Clostridium thermopalmarium*, and *Clostridium butyricum* are butyric acid producing bacteria. These results may lead to a conclusion for a promising future in terms of valorizing organic wastes with an overall biorefinery approach. However, the ideal conditions for long term production should be further optimized as the stability of anaerobic processes are dependent on many factors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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